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Superseding
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MILITARY SPECIFICATION

AMPLIFIER AM-203C/ARA-19

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of amplifier, designated Amplifier AM-203C/ARA-19.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing, and Marking of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

U. S. Air Force

55J13317	Amplifier AM-203C/ARA-19
55C13318	Wiring Diagram, AM-203C/ARA-19 Amplifier, Schematic

FED. SUP CLASS. 5826

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General.- The requirements, including materials, parts, and processes specified in Specification MIL-E-5400, are applicable as requirements of this specification, with the following additions or exceptions to Specification MIL-E-5400 as identified by the similarity of paragraph headings. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2.1 Samples required for part and material approvals.- Six samples of each nonstandard part shall be furnished the procuring activity.

3.2.2 Life.- The equipment shall provide at least 500 hours of life without repair or adjustment.

3.3 Wiring.- The wiring shall be in accordance with Drawing 55C13318.

3.4 Assembly.- The amplifier shall be assembled in accordance with Drawing 55J13317.

3.5 Calibration accuracy.- When Amplifier AM-203C/ARA-19 is used with Control panel C-685B/ARA-19 or Control panel C-423( )/A and Tuning drive PD-5( )/ARA-19, connected in the usual manner to Radio compass AN/ARN-6, the calibration accuracy shall be subject to the following limits:

Frequency band No.	Maximum error in setting
1	1.0 kc
2	1.5 kc
3	3.0 kc
4	6.0 kc

3.6 Identification of product.- For Navy procurement, nomenclature and nameplates shall be in accordance with MIL-N-18307.

3.7 Government-loaned property.- When provided for in the contract or purchase order, the following items will be furnished by the Government on loan to the contractor upon his request:

TABLE I

Government-loaned property

Item	Stock No.	Nomenclature	Quantity
1		Control panel C-685B/ARA-19 or Control panel C-423( )/A	1
2		Tuning drive PD-5( )/ARA-19	1
3	1600-214349860	Radio compass unit R-101( )/ARN-6	1
4	1600-298359383	Mounting MT-273( )/ARN-6	1
5	1600-212605034	Indicator ID-91( )/ARN-6	1
6	1600-322850505	Cord CG-121/ARN-6	1
7	1600-154433720	Cord CG-405/ARN-6	1
8	1600-203924710	Loop assembly AS-313( )/ARN-6	1
9		Handbook of Maintenance Instructions AN16-30ARN-6-3	1

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3.8 Workmanship.- All details of workmanship shall be in accordance with high-grade manufacturing practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of the amplifier shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the amplifier, to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests accomplished on amplifiers manufactured and submitted for acceptance under contract.

4.2 Preproduction tests.-

4.2.1 Sampling instructions.- Three amplifiers as specified in the contract will be tested for design approval by the procuring activity or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2 Tests.- The Preproduction tests shall consist of the Acceptance tests, the following tests, and any other tests deemed necessary by the procuring activity to determine that the amplifier meets all the requirements specified herein.

4.2.2.1 Environmental tests.- The Environmental tests shall be in accordance with the following procedures of Specification MIL-E-5272:

(a) Humidity tests: Procedure II.
(b) Salt spray tests: Procedure I.
(c) Vibration tests: Procedure XI.
(d) Fungus resistance tests: Procedure I.
(e) Explosion-proof (Aeronautical) tests: Procedure I.
(f) Temperature altitude test: Procedure I.
(g) Shock tests: Procedure II.

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests.- Each amplifier shall be subjected to the following tests.

4.3.1.1 The component parts of the amplifier shall be inspected to determine compliance with this specification with respect to materials and workmanship prior to assembly.

4.3.1.2 Mechanical inspection.- The amplifier shall be given a thorough mechanical inspection. Particular attention shall be paid to the quality and workmanship of the soldering, riveting, finishing, wire and wiring, tropicalization treatments, stamping, marking, and machine screw assemblage.

4.3.1.3 Electrical tests.- The amplifier shall be completely tested for correctness of wiring, continuity of all circuits, short circuits, and correct operation.

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4.3.2 Sampling tests.-

4.3.2.1 Life test.- At least one production Amplifier AM-203C/ARA-19 selected at random from each contract or order shall be subjected to 500 hours of continuous operation at room temperature. The equipment shall be subjected to a Life test when connected as a component of Remote tuning group AN/ARA-19 and the complete tuning group connected to Radio compass unit R-101( )ARN-6 with Mounting MT-273( )/ARN-6. At the end of the 500-hour test, the amplifier shall be inspected for defects resulting from the Life test. Any faults occurring which are not caused by accidental failure, as judged by the inspecting activity, shall be corrected in all production amplifiers. Particular attention shall be given to tube life and operation of the relays in AM-203C/ARA-19. A record shall be kept of all failures.

4.3.3 Rejection and retest.- Rejected amplifiers shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 General.- The amplifier shall be prepared for delivery in accordance with Specification MIL-E-17555.

5.2 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Stock No. or other identification number as specified
in the purchase document*
AMPLIFIER AM-203C/ARA-19
Specification MIL-A-6348D

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification as provided by the procuring activity.

6. NOTES

6.1 Intended use.- Amplifier AM-203C/ARA-19 is intended to be used in Remote tuning group AN/ARA-19 for remote tuning Radio compass AN/ARN-6.

6.2 Ordering data.- Procurement documents, contracts, and purchase orders should state the conditions for the following.

6.2.1 Preproduction tests.- It is expected that the contract or purchase order will specify that three amplifiers will be required as first articles (or preproduction samples) and that these items will be subjected to the Preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Approved model.- After acceptance of the three first articles by the procuring activity, one shall be shipped to the contractor for use as an approved model by the Inspector for the duration of the contract.

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Custodians:
Navy - Bureau of Aeronautics
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